

ANURAK NAMGAT

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SUMMARY

I am an Industrial Physics senior at KMITL with experience designing and developing systems across backend, database, and frontend.

Currently developing a multiplayer simulation game, where I manage player state synchronization, implement JWT-based authentication, and ensure data integrity. I also build frontend features for real-time interactions and an admin dashboard.

Interested in scalable systems, handling concurrent users, and solving problems in distributed state and real-time processing.

SKILLS

- Programming Languages: TypeScript, JavaScript, Python, Julia, C/C++
- Backend: NestJS, Node.js, REST API, WebSocket (Socket.IO)
- Frontend: React, Phaser.js, Vite
- Database & ORM: PostgreSQL, Redis, Prisma
- Infrastructure & Tools: Docker, Nginx, PM2, Linux, Git
- Communication: English (Intermediate), Thai (Native)

PROJECT & INTERNSHIP EXPERIENCE

Finix — Real-time Multiplayer Financial Simulation Game	Bangkok, Thailand
<i>Full-stack Developer KMITL Research Project</i>	May 2025 - Present
- Developed a real-time multiplayer card-based financial game (2–4 players) using NestJS, React, and Socket.IO, and deployed the system on an Ubuntu production server. - Designed a 24-table PostgreSQL schema via Prisma ORM to manage 576 game cards, 15 careers, and financial data with Decimal(18,2) precision, including 13 database migrations. - Built 47+ REST endpoints and 9 WebSocket events across 2 namespaces to handle turn-based gameplay, dice rolling, banking, and player-to-player asset exchange. - Solved cross-instance state inconsistency in a PM2 cluster by integrating a Redis Pub/Sub adapter, and implemented an auto-skip turn system (45s timeout) and session cleanup. - Implemented a 7-dimension financial scoring algorithm from an academic paper, built an 8-page React admin dashboard, and configured JWT authentication, RBAC, and Redis-backed rate limiting.	
CP ALL Public Company Limited	Bangkok, Thailand
<i>Data Development Intern</i>	April 2024 - June 2024
- Developed an interactive sales dashboard using Python (Streamlit) and Plotly to visualize store performance and support business insights across 7-Eleven branches - Optimized data pipeline for a 685,000-row compressed dataset (202 MB) by implementing Streamlit caching (@st.cache_data), reducing data load time from ~3.6s to ~0.07s — a 98% reduction in load time and significantly improving dashboard responsiveness	

Micro-Defect Detection on Metal Surfaces (Senior Project)

Bangkok, Thailand

Algorithm Developer Tech Stack: Julia, Digital Holography, FFT, ASM, Signal & Image Processing

August 2025 - Present

- Engineered a Julia-based algorithmic pipeline combining the Angular Spectrum Method (ASM), auto-focusing, and quality-guided phase unwrapping to reconstruct 3D surface topography from 4K digital holograms.
- Applied 2D least-squares quadratic polynomial fitting and edge-preserving filters to eliminate lens aberration, achieving an RMS noise floor of 8.45 nm and enabling precise defect detection at a 22.08 nm threshold.
- Extracted nanometer-scale 3D measurements without traditional microscopes, successfully detecting 229 simultaneous micro-defects and profiling surface anomalies ranging from -90.50 nm to 95.52 nm.

IoT Smart Vehicle Control System

Bangkok, Thailand

Embedded Systems Developer | KMITL Physics Lab (Microcontroller)

October 2024 - November 2024

- Engineered an autonomous vehicle using Arduino C++ and a 5-state Finite State Machine (FSM) to manage mission phases, integrating IR, ultrasonic, DHT11, LDR, and I2C LCD components
- Implemented dynamic navigation logic encompassing line-following, tunnel navigation, and an obstacle avoidance system that triggers pre-timed maneuver sequences at <22 cm detection
- Developed environmental responsiveness, displaying real-time temperature/humidity data on an I2C LCD and executing automated state transitions based on specific thresholds (>30°C)

EDUCATION

Bachelor of Science (Industrial Physics)

Bangkok, Thailand

King Mongkut's Institute of Technology Ladkrabang (KMITL)

August 2022 - Present

Faculty of Science, 3.23 GPAX

Expected graduation: May 2026

CERTIFICATIONS

True Digital Academy : Data Analytics Specialize Track | May 2024

True Digital Academy : User Experience Design Kickstarter | Jan 2025

42 Bangkok : Discovery Piscine Basic Web Development (Achieved maximum level 13.57/13.57) | Feb 2026